(d) With the help of a neat sketch, explain the principle of flash welding with its merits and demerits. 7

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5. (a) Define sag and swell. 2

(b) Describe principle of Buck Boost control of servo controlled voltage stabilizer with the help of a neat sketch. 7

(c) Describe the features of line interactive UPS and stand by power supply UPS system. 7

Or

A 2 V lead acid cell is discharged at a uniform rate of 35 A for 5 hours at an average emf of 1.95 V. If it is then charged at a rate of 45 Amp for 4 hours to restore its original voltage, determine :

(i) A-H efficiency

(ii) W-H efficiency

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(d) How is PRRR method helpful to reduce power line disturbances ? 7

Or

Define the following terms regarding power line disturbances :

(i) Brown-out

(ii) Black-out

(iii) Earth potential rise

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B. E. (Eighth Semester)
EXAMINATION, April-May, 2015
(Old Scheme)
(Branch : Et & T)
INDUSTRIAL AND POWER ELECTRONICS

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : (i) Attempt all questions.

(ii) Part (a) of each question is compulsory.

(iii) Attempt any *two* parts from (b), (c) and (d).

(iv) Sketch circuit diagram and waveforms with proper attention, wherever required.

1. (a) Define line regulation and load regulation. 2

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(b) Describe unsuitability of unregulated power supply. Define source regulation factor, output resistance and temperature coefficient of voltage for stabilization of power supply. 7

(c) Explain the Zener diode voltage regulator and analyze the operation under the condition of : 7

(i) Fixed *vs.* variable R_L

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(ii) Variable *vs.* fixed R_L

(d) Write short notes on any *two* of the following : 7

(i) Dual tracking regulator

(ii) Switch mode power supply

(iii) Op-Amp shunt voltage regulator

2. (a) Justify the statement "Higher is the gate current, lower is the forward breakover voltage". 2

(b) A relaxation oscillator using an UJT is to be designed for triggering an SCR.

The UJT has the following data intrinsic stand off ratio = 0.72,

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Peak current = 0.6 mA, Peak voltage = 18.0 V

Valley current = 2.5 mA, Valley voltage = 1.0 V

R_{BB} = 5 kΩ, Normal leakage current with emitter open = 4.2 mA.

The firing frequency is 2 kHz.

For C = 0.04 μF

Compute the value of R, R_1 and R_2 and V_{BB}. 7

(c) Explain how SCR's suffer from unequal voltage distribution during their turn on and turn off process with the help of waveforms. 7

(d) What is the basic difference between load commutation and line commutation ? 7

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Or

With the help of neat sketch and wave forms explain parallel capacitor commutation. 7

3. (a) State the difference between circulating and non-circulating current mode dual converter. 2

(b) With the help of a neat sketch and wave forms, explain the principle and working of 120°, conduction mode 3-phase bridge inverter. 7

(c) With the help of neat sketch and waveforms, explain the working of Jone's chopper. 7

(d) What is the basic difference between A. C. voltage controller and cycloconverter. 7

Or

With the help of a neat sketch and waveforms explain the single phase to single phase cycloconverter in bridge type configuration. 7

4. (a) Define skin effect. 2

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(b) Explain the principle of induction heating and find out expression of total power developed at the surface in induction heating. 7

(c) Write the features of dielectric heating. 7

Or

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A workpiece of dielectric material with size 20 cm × 15 cm × 1 cm is heated using frequency of 20 MHz and applied voltage of 750 V. If relative permittivity of material is 3 and power factor is 0.4, calculate power developed per unit volume. 7